

Signals and Systems

Chapter 2

Time-Domain Representations of Linear Time-Invariant Systems

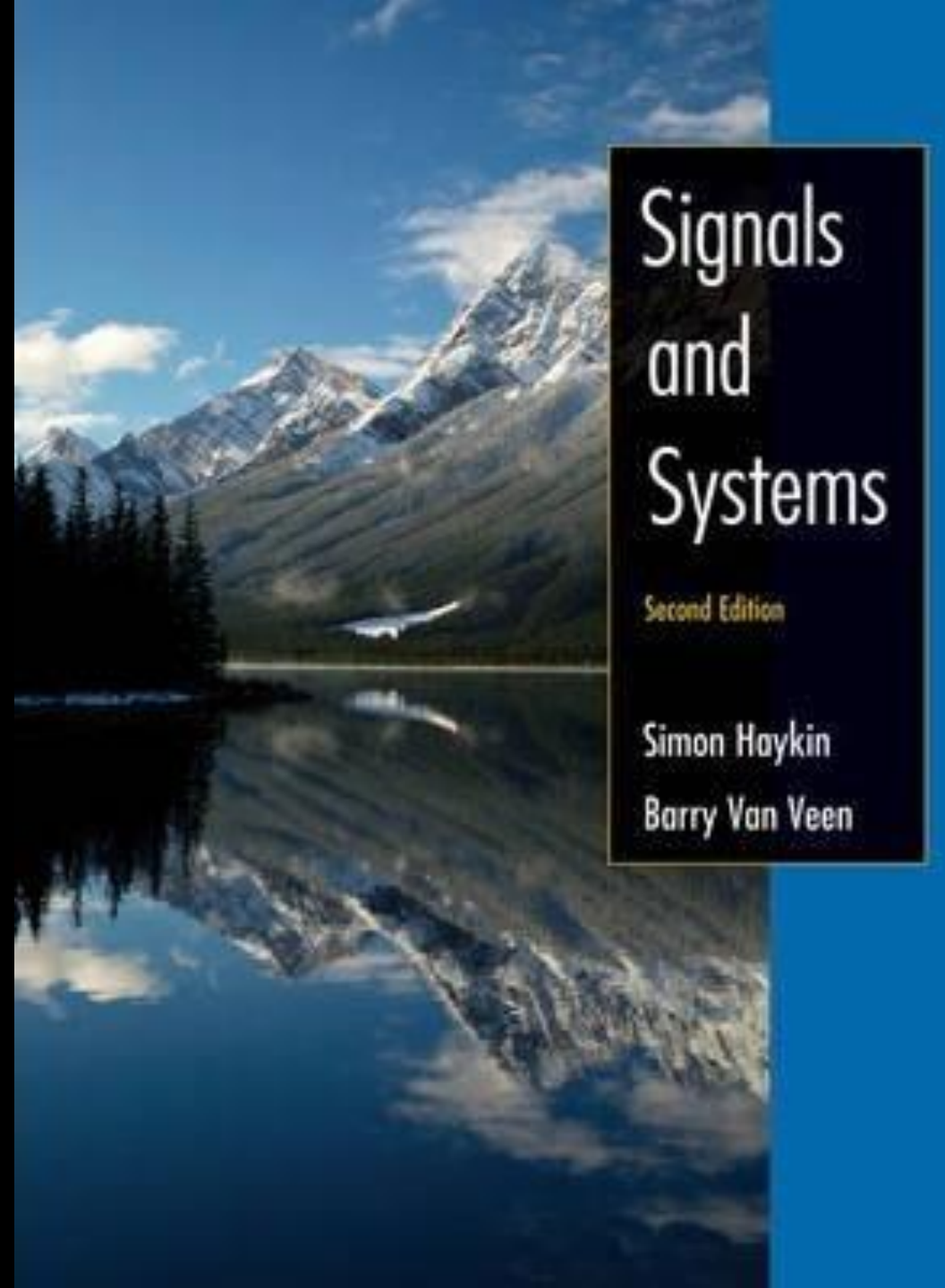
Convolution 2

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Kaynaklar :

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Chapter 2

Time-Domain Representations of Linear Time-Invariant Systems

Objectives of this chapter

1. Introduction
2. The Convolution Sum
3. Convolution Sum Evaluation Procedure
4. The Convolution Integral
5. Convolution Integral Evaluation Procedure
6. Interconnections of LTI Systems
7. Relations between LTI System Properties and the Impulse Response
8. Step Response
9. Differential and Difference Equation Representations of LTI Systems
10. Solving Differential and Difference Equations
11. Characteristics of Systems Described by Differential and Difference Equations
12. Block Diagram Representations
13. State-Variable Descriptions of LTI Systems
14. Exploring Concepts with MATLAB
15. Summary



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2.4 The Convolution Integral

- The output of a continuous-time LTI system may be determined from the input and the system's impulse response.
- We first express a continuous-time signal as weighted superposition of time-shifted impulses:

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau \quad (2.9)$$

- Where superposition is the sum
- Let the operator H denote the system to which the input $x(t)$ is applied. The system output in response to a general input expressed as the weighted superposition in Eq. (2.9):

$$y(t) = H\{x(t)\} = H\left\{\int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau\right\} = \int_{-\infty}^{\infty} x(\tau) H\{\delta(t - \tau)\} d\tau$$

- Next, impulse response $h(t) = H\{\delta(t)\}$ is defined as the output of the system in response to a unit impulse input.

$$H\{\delta(t - \tau)\} = h(t - \tau) \quad (2.11)$$

- That is, time invariance implies that a time-shifted impulse input generates a time-shifted impulse response output, as shown in Fig. 2.9.

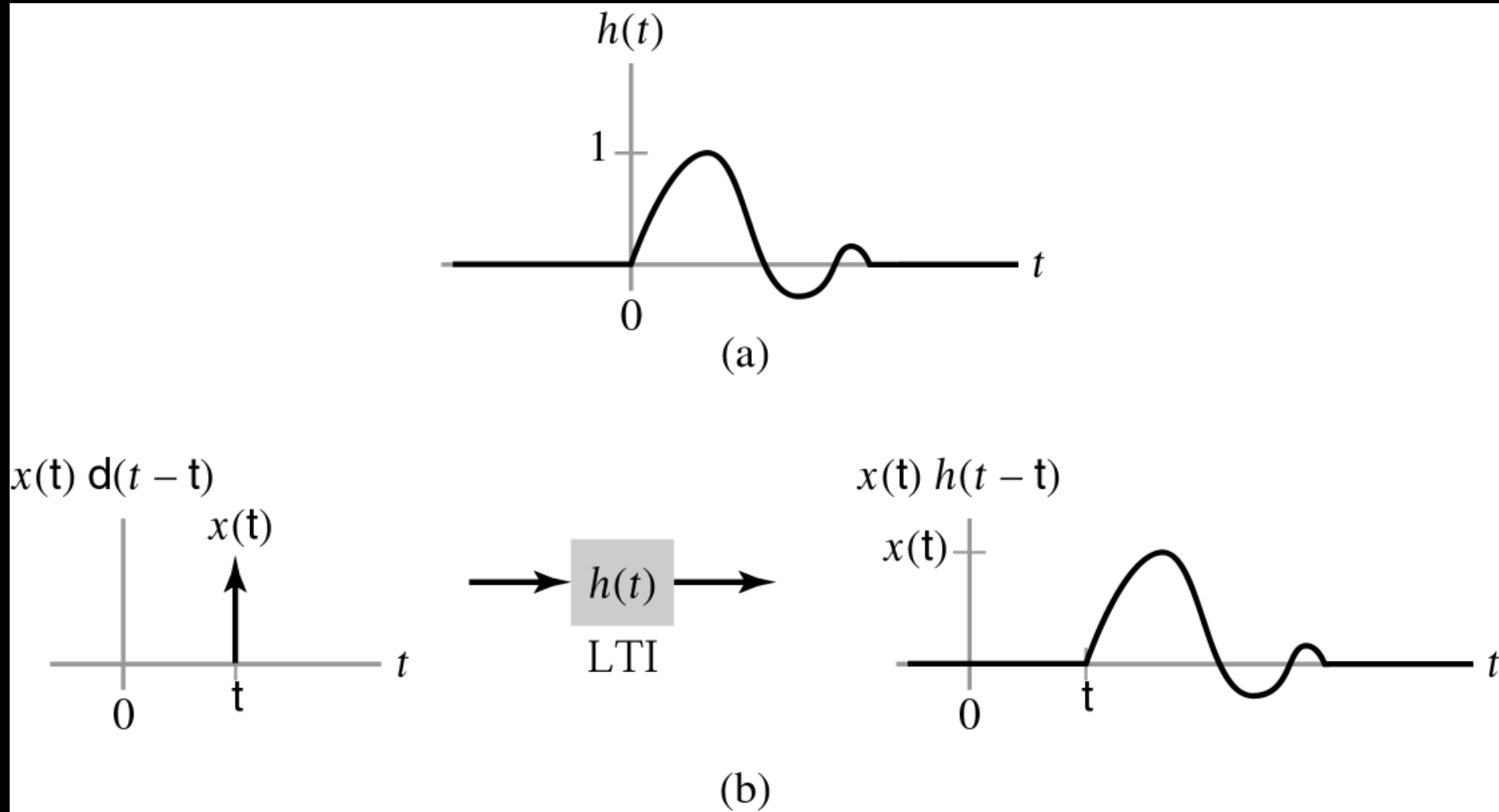


Figure 2.9 (p. 115)

(a) Impulse response of an LTI system H . (b) The output of an LTI system to a time-shifted and amplitude-scaled impulse is a time-shifted and amplitude-scaled impulse response.

- we see that the output of an LTI system in response to an input of the form of Eq. (2.9) may be expressed as

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \quad (2.12)$$

- The output $y(t)$ is given as a weighted superposition of impulse responses time shifted by τ . Equation (2.12) is termed the convolution integral and is also denoted by the symbol $*$; that is,

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$

2.5 Convolution Integral Evaluation Procedure

- The convolution integral of Eq. (2.12) is expressed as

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau \quad (2.12)$$

Defining integrand as intermediate signal $w_t(\tau) = x(\tau)h(t - \tau)$, (2.12) is written as

$$y(t) = \int_{-\infty}^{\infty} w_t(\tau)d\tau \quad (2.14)$$

EXAMPLE 2.6 REFLECT-AND-SHIFT CONVOLUTION EVALUATION Evaluate the convolution integral for a system with input $x(t)$ and impulse response $h(t)$, respectively, given by

$$x(t) = u(t - 1) - u(t - 3)$$

and

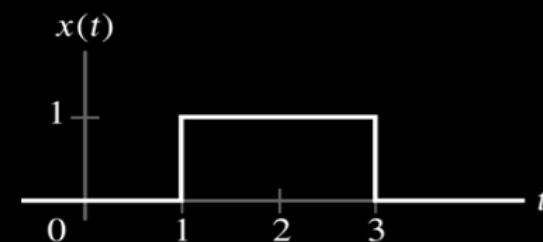
$$h(t) = u(t) - u(t - 2),$$

as depicted in Fig. 2.10.

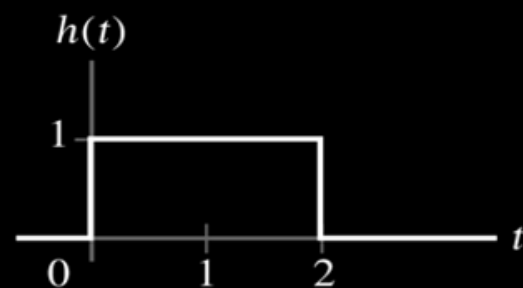
Figure 2.10 (p. 117)

Input signal and LTI system impulse response for Example 2.6.

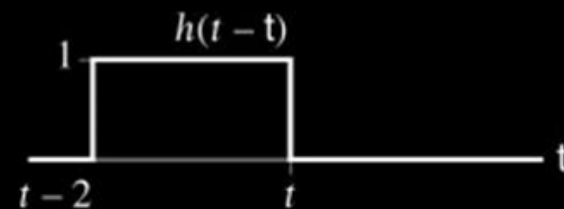
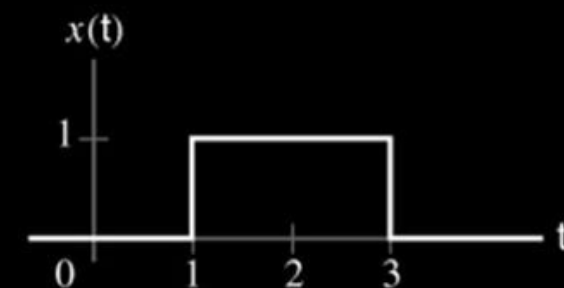
$$x(t) = u(t - 1) - u(t - 3)$$



$$h(t) = u(t) - u(t - 2)$$



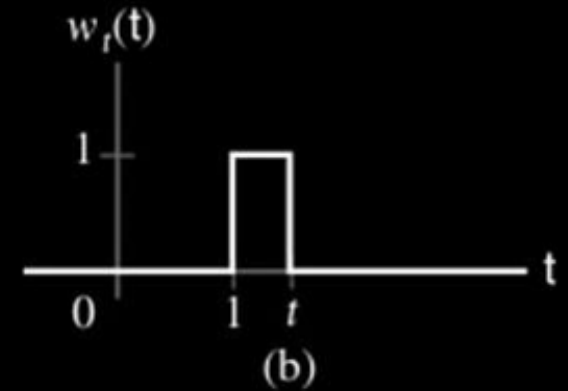
Solution: To evaluate the convolution integral, we first graph $h(t - \tau)$ beneath the graph of $x(\tau)$, as shown in Fig. 2.11(a). Next, we identify the intervals of time shifts for which the mathematical representation of $w_i(\tau)$ does not change, beginning with t large and negative. Provided that $t < 1$, we have $w_i(\tau) = 0$, since there are no values τ for which both $x(\tau)$ and $h(t - \tau)$ are nonzero. Hence, the first interval of time shifts is $t < 1$.



(a)

Note that at $t = 1$ the right edge of $h(t - \tau)$ coincides with the left edge of $x(\tau)$. Therefore, as we increase the time shift t beyond 1, we have

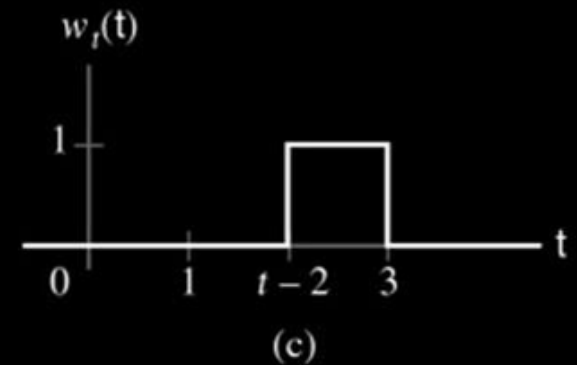
$$w_t(\tau) = \begin{cases} 1, & 1 < \tau < t \\ 0, & \text{otherwise} \end{cases}.$$



This representation for $w_t(\tau)$ is depicted in Fig. 2.11(b). It does not change until $t > 3$, at which point both edges of $h(t - \tau)$ pass through the edges of $x(\tau)$. The second interval of time shifts t is thus $1 \leq t < 3$.

As we increase the time shift t beyond 3, we have

$$w_t(\tau) = \begin{cases} 1, & t - 2 < \tau < 3 \\ 0, & \text{otherwise} \end{cases},$$

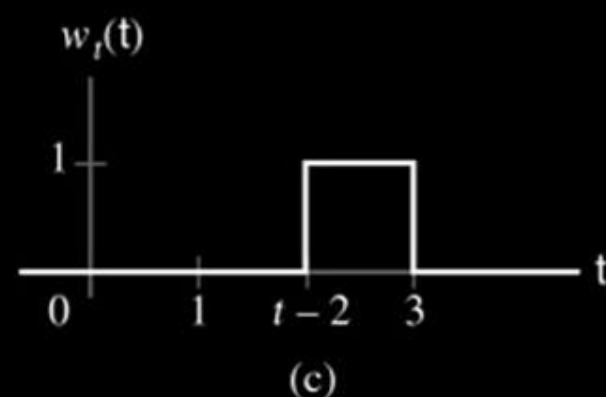
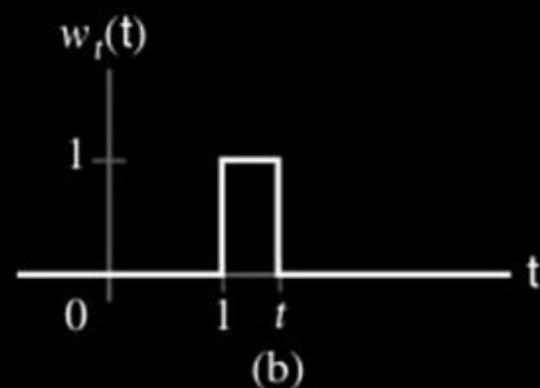


as depicted in Fig. 2.11(c). This mathematical representation for $w_t(\tau)$ does not change until $t = 5$; thus, the third interval of time shifts is $3 \leq t < 5$.

At $t = 5$, the left edge of $h(t - \tau)$ passes through the right edge of $x(\tau)$, and $w_t(\tau)$ becomes zero. As we continue to increase t beyond 5, $w_t(\tau)$ remains zero, since there are no values τ for which both $x(\tau)$ and $h(t - \tau)$ are nonzero. Hence, the final interval of shifts is $t \geq 5$.

We now determine the output $y(t)$ for each of these four intervals of time shifts by integrating $w_t(\tau)$ over τ (i.e., finding the area under $w_t(\tau)$):

- For $t < 1$ and $t > 5$, we have $y(t) = 0$, since $w_t(\tau)$ is zero.
- For the second interval, $1 \leq t < 3$, the area under $w_t(\tau)$ shown in Fig. 2.11(b) is $y(t) = t - 1$.
- For $3 \leq t < 5$, the area under $w_t(\tau)$ shown in Fig. 2.11(c) is $y(t) = 3 - (t - 2)$.



Combining the solutions for each interval of time shifts gives the output

$$y(t) = \begin{cases} 0, & t < 1 \\ t - 1, & 1 \leq t < 3 \\ 5 - t, & 3 \leq t < 5 \\ 0, & t \geq 5 \end{cases},$$

as shown in Fig. 2.11(d).

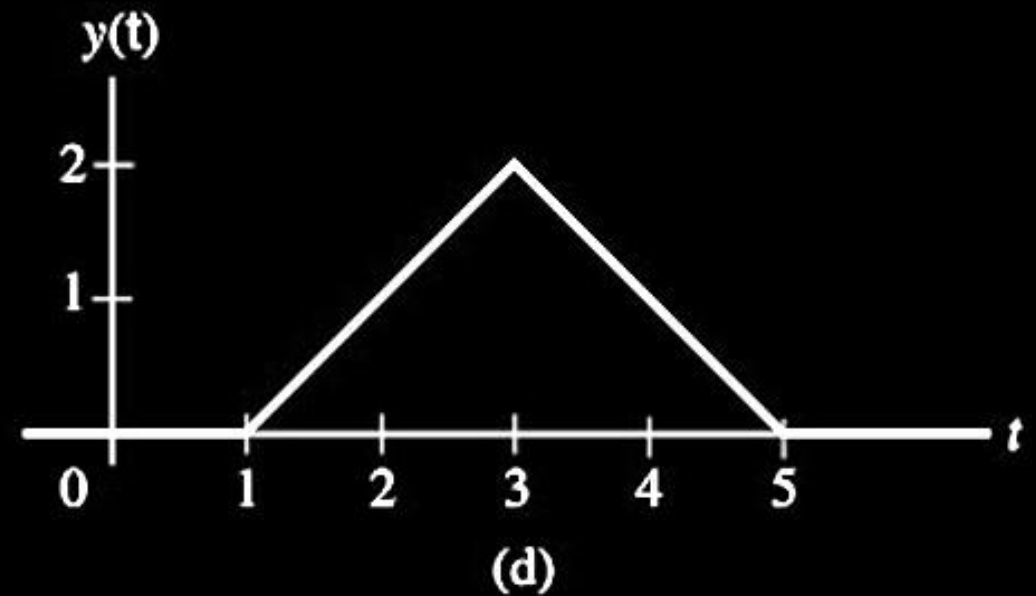


Figure 2.11 (d) The system output $y(t)$.

Figure 2.11 (p. 118)

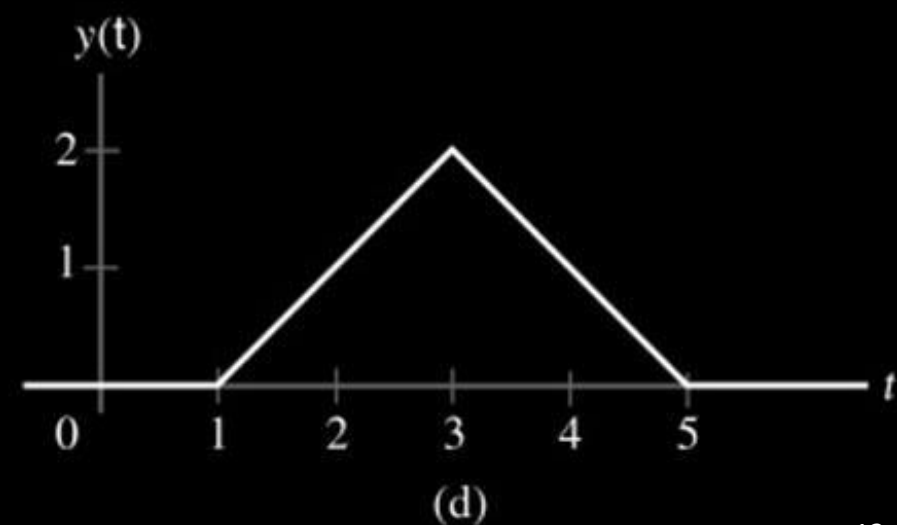
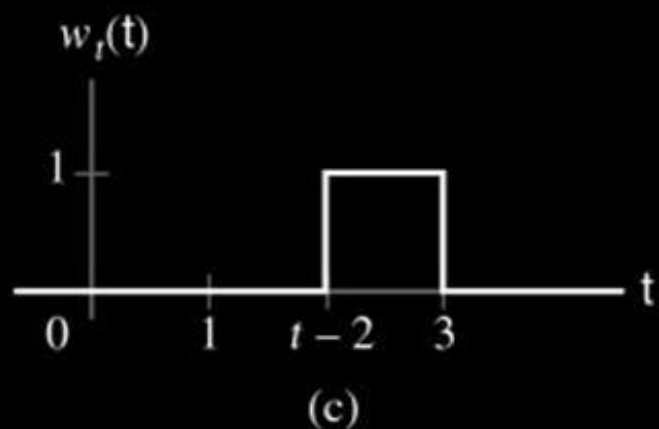
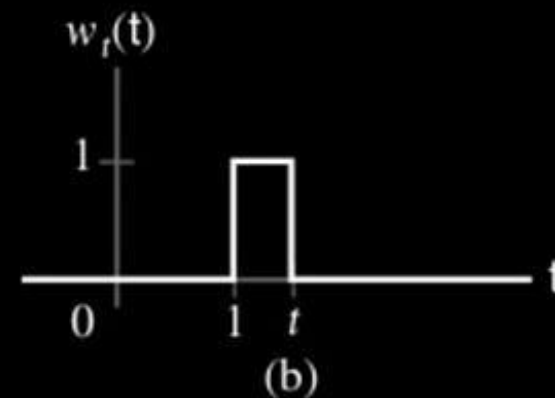
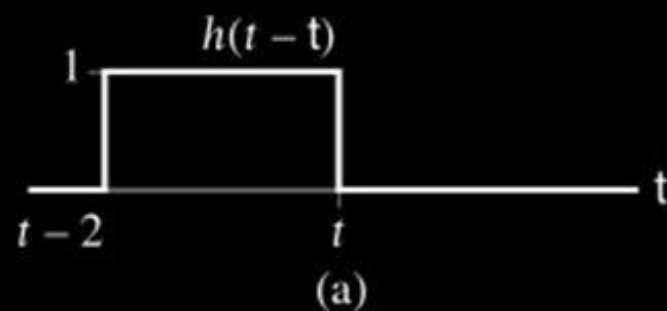
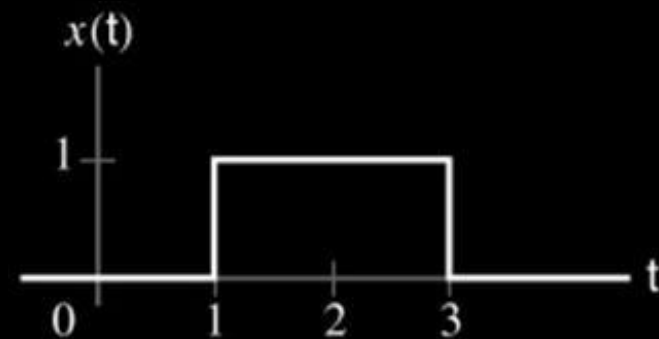
Evaluation of the convolution integral for Example 2.6.

(a) The input $x(\tau)$ depicted above the reflected and time-shifted impulse response.

(b) The product signal $w_t(\tau)$ for $1 \leq t < 3$.

(c) The product signal $w_t(\tau)$ for $3 \leq t < 5$.

(d) The system output $y(t)$.



EXAMPLE 2.7 RC CIRCUIT OUTPUT Consider the RC circuit depicted in Fig. 2.12, and assume that the circuit's time constant is $RC = 1$ s. Example 1.21 shows that the impulse response of this circuit is

$$h(t) = e^{-t}u(t).$$

Use convolution to determine the voltage across the capacitor, $y(t)$, resulting from an input voltage $x(t) = u(t) - u(t - 2)$.

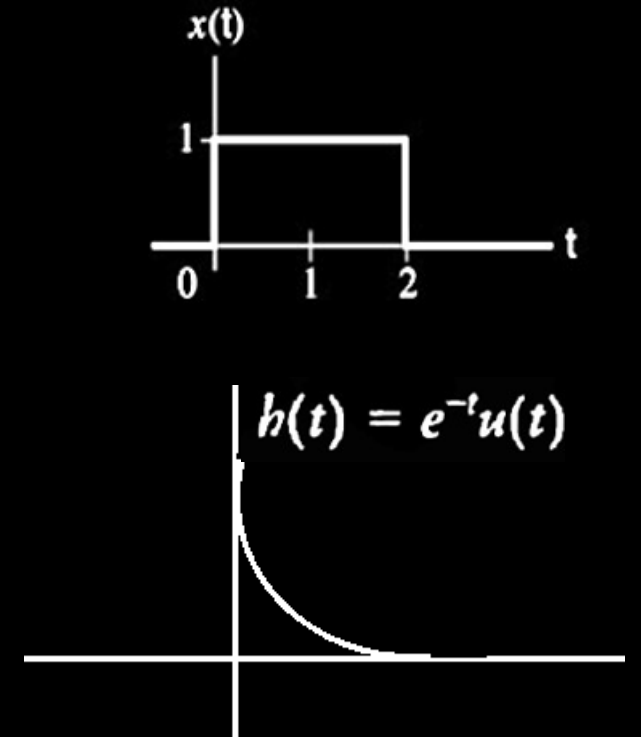
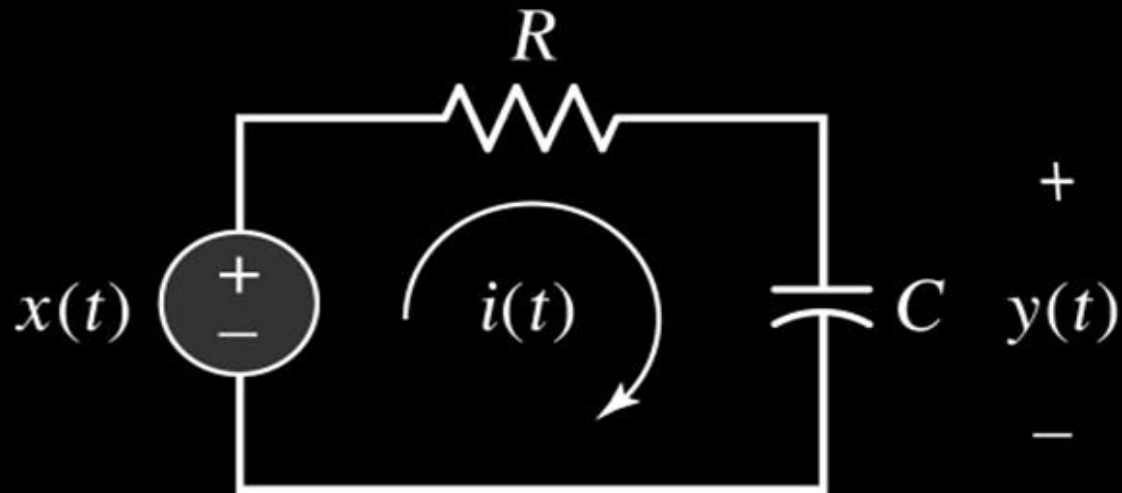
Solution: The circuit is linear and time invariant, so the output is the convolution of the input and the impulse response. That is, $y(t) = x(t) * h(t)$. Our intuition from circuit analysis indicates that the capacitor should charge toward the supply voltage in an exponential fashion beginning at time $t = 0$, when the voltage source is turned on, and then, at time $t = 2$, when the voltage source is turned off, start to discharge exponentially.

To verify our intuition using the convolution integral, we first graph $x(\tau)$ and $h(t - \tau)$ as functions of the independent variable τ . We see from Fig. 2.13(a) that

$$x(\tau) = \begin{cases} 1, & 0 < \tau < 2 \\ 0, & \text{otherwise} \end{cases}$$

Figure 2.12 (p. 119)

RC circuit system with the voltage source $x(t)$ as input and the voltage measured across the capacitor $y(t)$, as output.



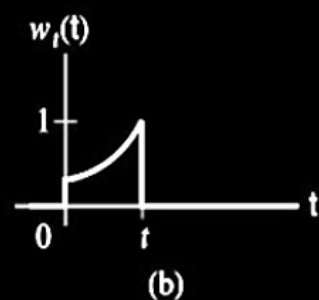
and

$$h(t - \tau) = e^{-(t-\tau)}u(t - \tau) = \begin{cases} e^{-(t-\tau)}, & \tau < t \\ 0, & \text{otherwise} \end{cases}$$

Now we identify the intervals of time shifts t for which the mathematical representation of $w_t(\tau)$ does not change. Begin with t large and negative. Provided that $t < 0$, we have $w_t(\tau) = 0$, since there are no values τ for which $x(\tau)$ and $h(t - \tau)$ are both nonzero. Hence, the first interval of shifts is $t < 0$.

Note that at $t = 0$ the right edge of $h(t - \tau)$ intersects the left edge of $x(\tau)$. For $t > 0$,

$$w_t(\tau) = \begin{cases} e^{-(t-\tau)}, & 0 < \tau < t \\ 0, & \text{otherwise} \end{cases}$$



This representation for $w_t(\tau)$ is depicted in Fig. 2.13(b). It does not change its mathematical representation until $t > 2$, at which point the right edge of $h(t - \tau)$ passes through the right edge of $x(\tau)$. The second interval of shifts t is thus $0 \leq t < 2$.

For $t \geq 2$, we have a third representation of $w_t(\tau)$, which is written as

$$w_t(\tau) = \begin{cases} e^{-(t-\tau)}, & 0 < \tau < 2 \\ 0, & \text{otherwise} \end{cases}.$$

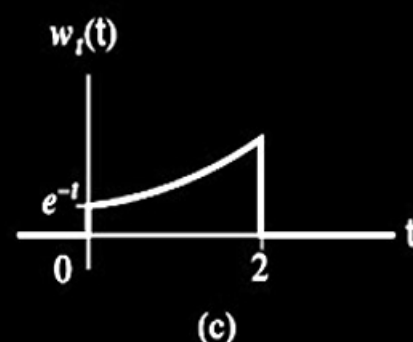


Figure 2.13(c) depicts $w_t(\tau)$ for this third interval of time shifts, $t \geq 2$.

We now determine the output $y(t)$ for each of the three intervals of time shifts by integrating $w_t(\tau)$ from $\tau = -\infty$ to $\tau = \infty$. Starting with the first interval, $t < 0$, we have $w_t(\tau) = 0$, and thus, $y(t) = 0$. For the second interval, $0 \leq t < 2$,

$$\begin{aligned} y(t) &= \int_0^t e^{-(t-\tau)} d\tau \\ &= e^{-t} (e^{\tau} \big|_0^t) \\ &= 1 - e^{-t}. \end{aligned}$$

For the third interval, $t \geq 2$, we have

$$\begin{aligned}y(t) &= \int_0^2 e^{-(t-\tau)} d\tau \\&= e^{-t}(e^{\tau}|_0^2) \\&= (e^2 - 1)e^{-t}.\end{aligned}$$

Combining the solutions for the three intervals of time shifts gives the output

$$y(t) = \begin{cases} 0, & t < 0 \\ 1 - e^{-t}, & 0 \leq t < 2, \\ (e^2 - 1)e^{-t}, & t \geq 2 \end{cases}$$

as depicted in Fig. 2.13(d). This result agrees with our intuition from circuit analysis.

Figure 2.13 (p. 120)

Evaluation of the convolution integral for Example 2.7.

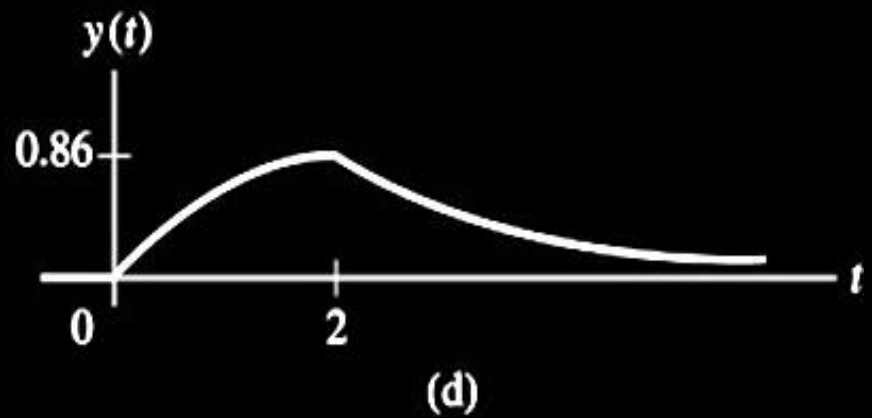
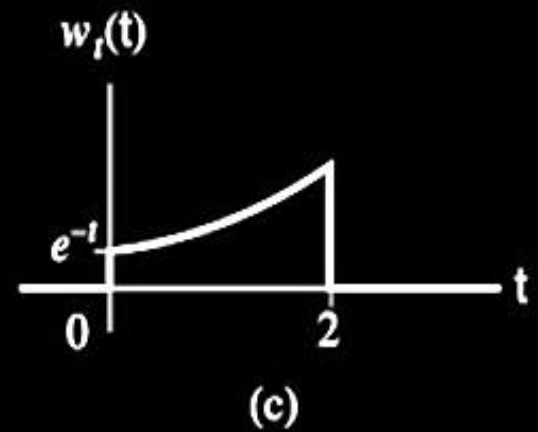
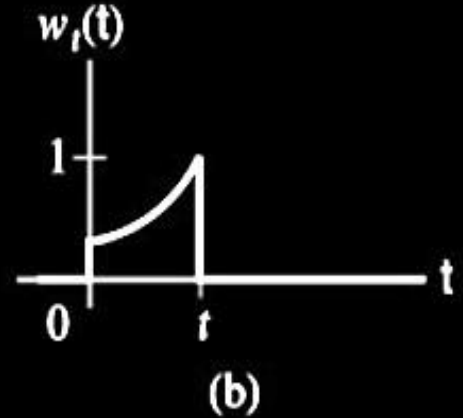
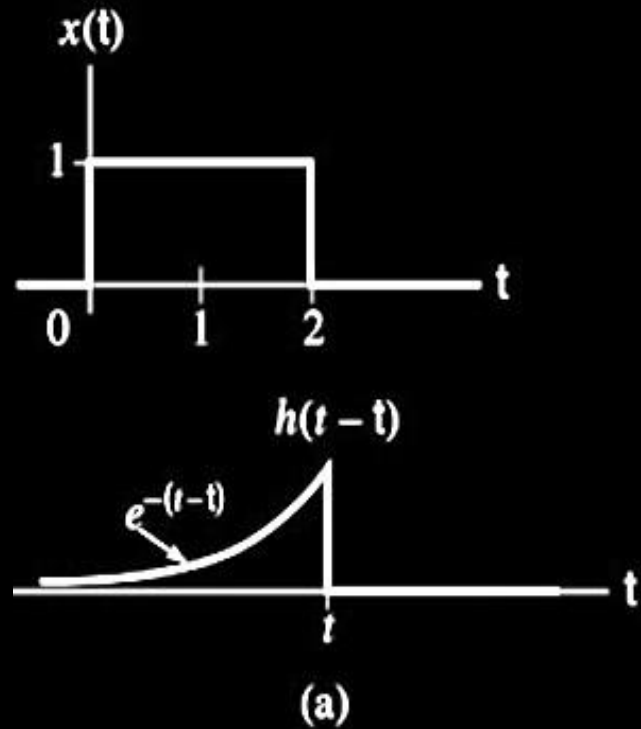
(a) The input $x(\tau)$ superimposed over the reflected and time-shifted

impulse response $h(t - \tau)$, depicted as a function of τ .

(b) The product signal $w_t(\tau)$ for $0 \leq t < 2$.

(c) The product signal $w_t(\tau)$ for $t \geq 2$.

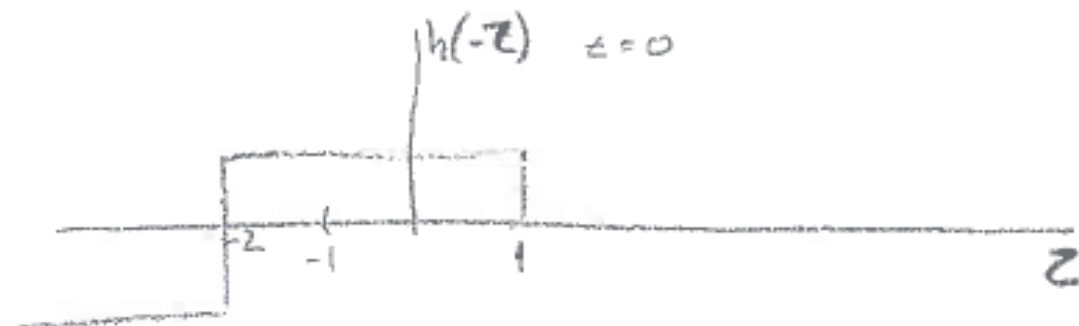
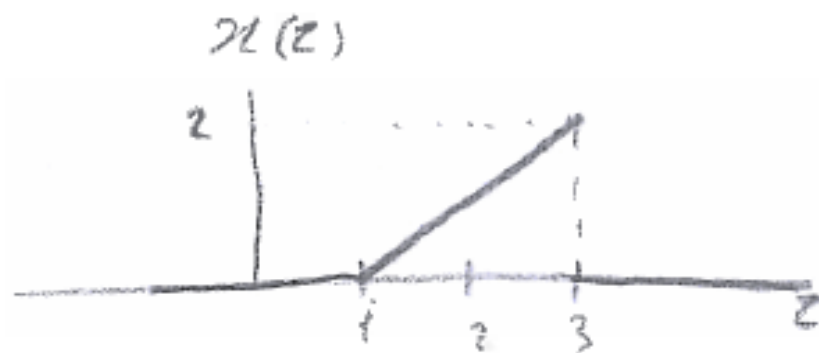
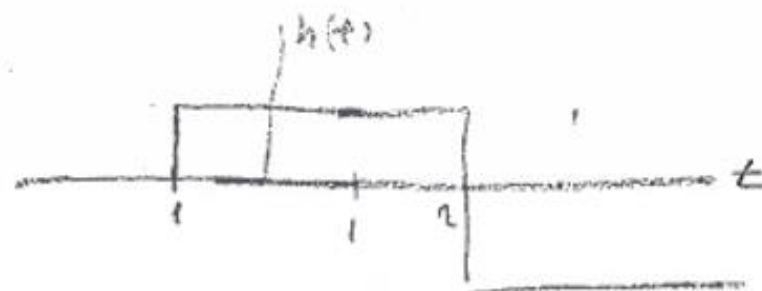
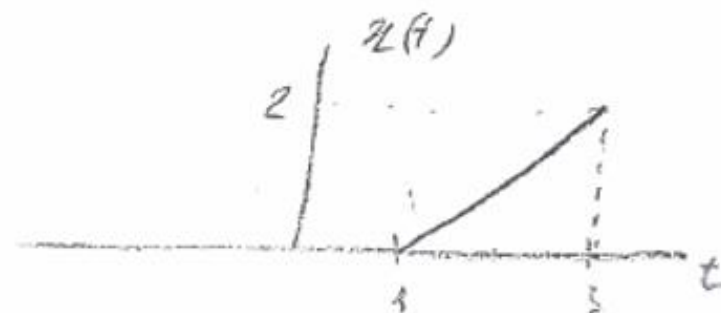
(d) The system output $y(t)$.



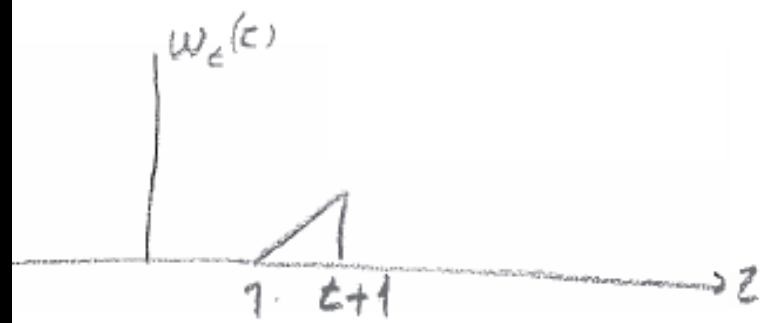
Exp 2.8 $y(t)=?$

$$x(t) = (t-1) [u(t-1) - u(t-3)]$$

$$h(t) = u(t+1) - 2u(t-2)$$



between $0 \leq t < 2$

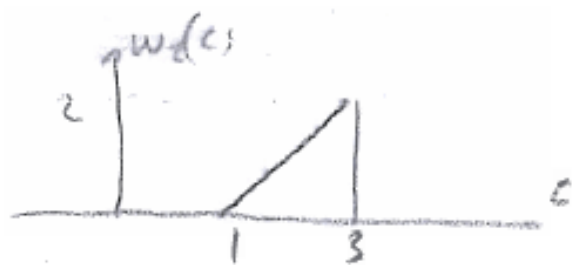


$$y(t) = \int_1^{t+1} (z-1) dz = \left. \frac{z^2}{2} - z \right|_1^{t+1} = \frac{(t+1)^2}{2} - (t+1) - \frac{1}{2} + 1$$

$$= \frac{t^2}{2} + t + \frac{1}{2} - t - 1 + \frac{1}{2}$$

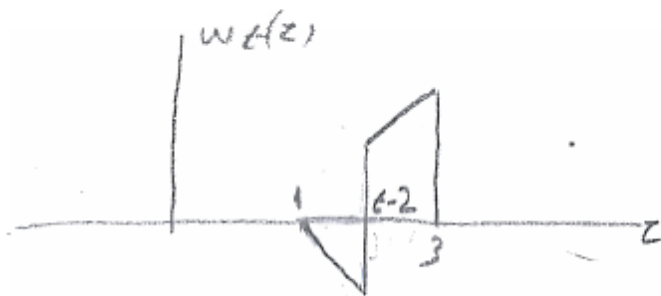
$$= \frac{t^2}{2}$$

between $2 \leq t < 3$



$$y(t) = 2$$

between $3 \leq t < 5$



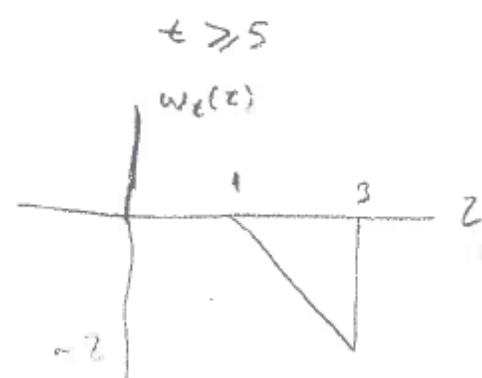
$$y(t) = \int_1^{t-2} (-z+1) dz + \int_{t-2}^3 (z-1) dz = -\frac{z^2}{2} + z \Big|_1^{t-2} + \frac{z^2}{2} - z \Big|_{t-2}^3$$

$$= -\frac{(t-2)^2}{2} + (t-2) + \frac{1}{2} - 1 + \frac{9}{2} - 3 - \frac{(t-2)^2}{2} + (t-2)$$

$$= -\left(\frac{t^2}{2} - 2t + 2\right) \cdot 2 + 2t - 4 + 1$$

$$= -t^2 + 4t + 2t - 7$$

$$= -t^2 + 6t - 7$$



$$y(t) = -2$$

$$y(t) = \begin{cases} 0 & t < 0 \\ t/2 & 0 \leq t < 2 \\ 2 & 2 \leq t < 3 \\ -t^2 + 6t - 7 & 3 \leq t < 5 \\ -2 & t \geq 5 \end{cases}$$

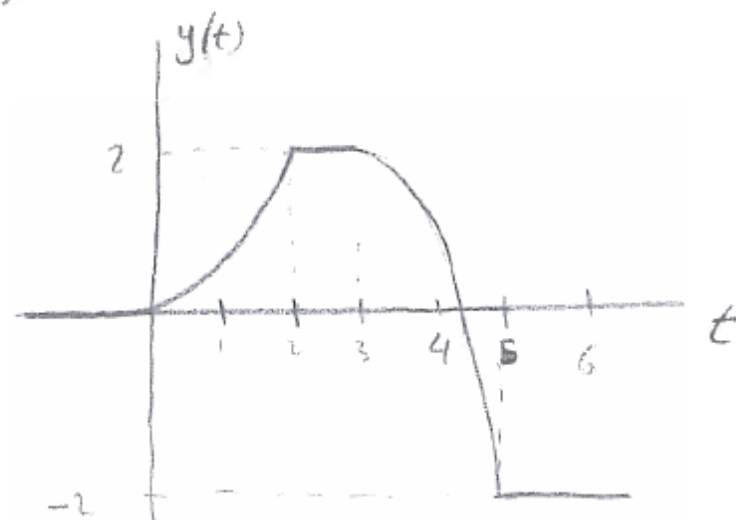


Figure 2.14 (p. 121)

Evaluation of the convolution integral for Example 2.8.

a) The input $x(\tau)$ superimposed on the reflected and time-shifted impulse response $h(t - \tau)$, depicted as a function of τ .

b) The product signal $w_t(\tau)$ for $0 \leq t < 2$.

c) The product signal $w_t(\tau)$ for $2 \leq t < 3$.

d) The product signal $w_t(\tau)$ for $3 \leq t < 5$.

e) The product signal $w_t(\tau)$ for $t \geq 5$. The system output $y(t)$.

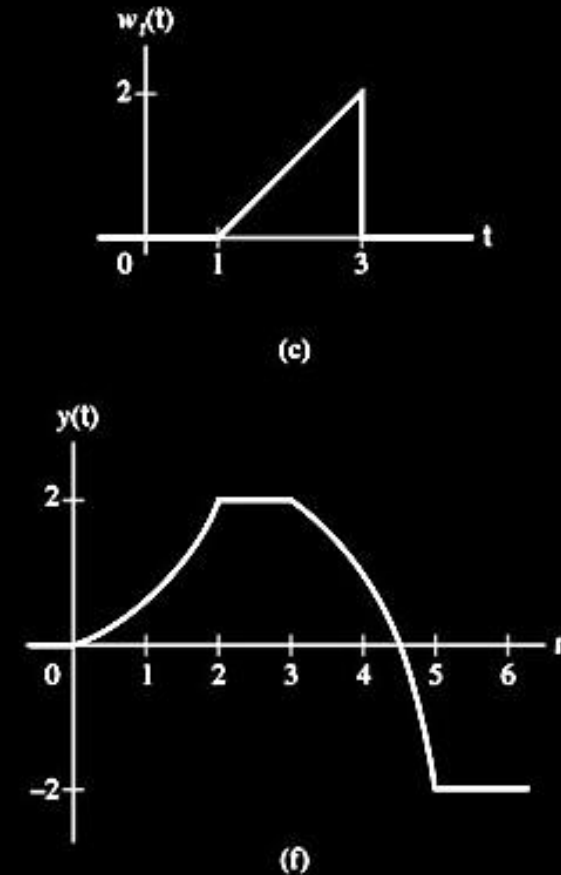
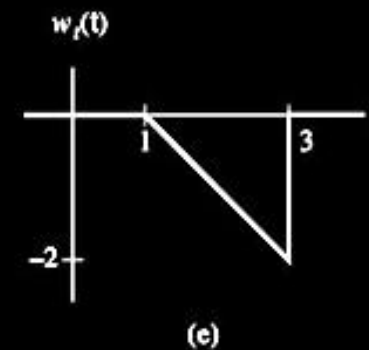
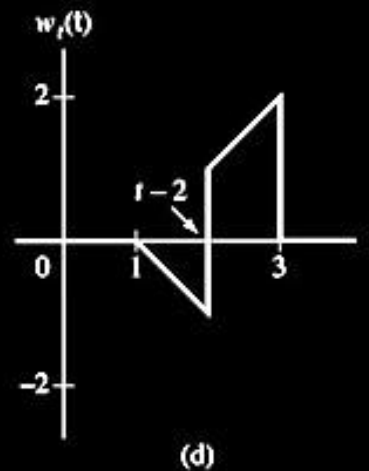
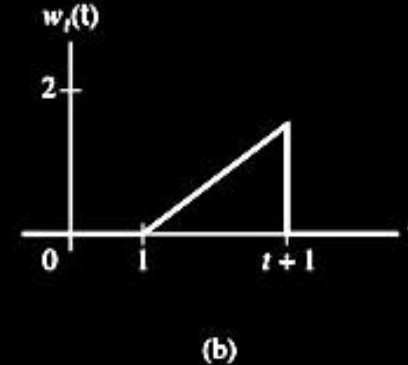
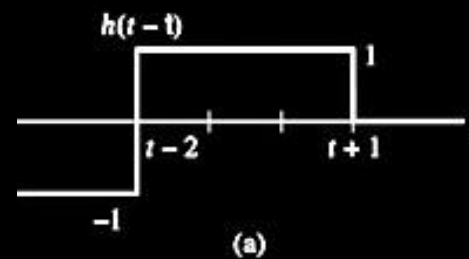
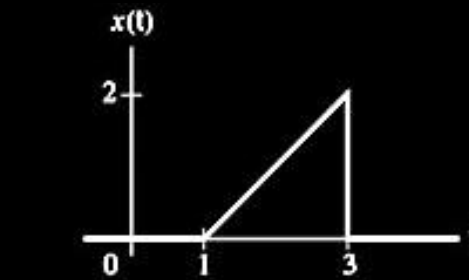
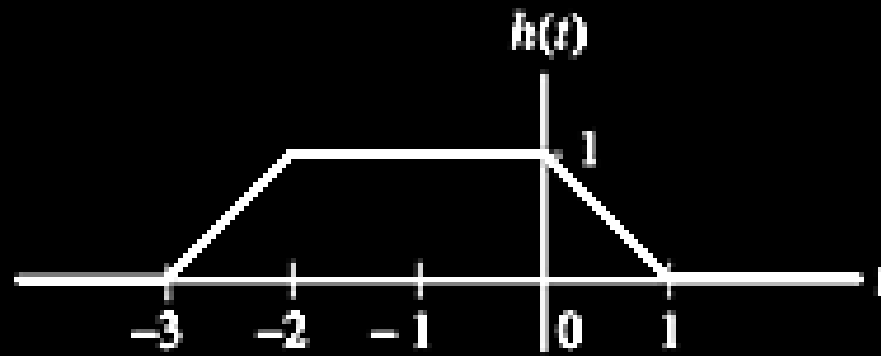
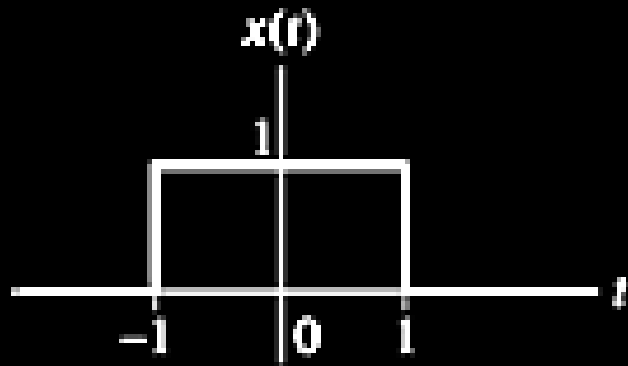
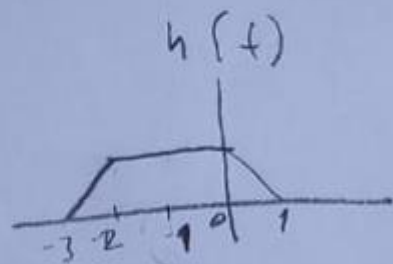
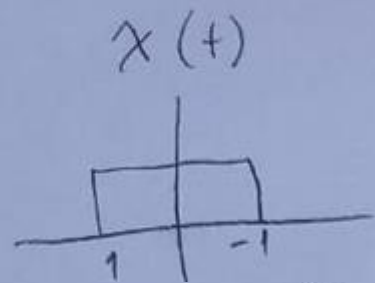


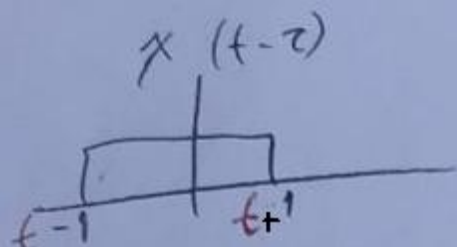
Figure 2.15 (p 123)

Homework





$$y(t) = \int_{-\infty}^{\infty} x(t-\tau) h(\tau) d\tau$$



$t < -4 \quad y(t) = 0$

$-4 \leq t < -3$ i.e. in;

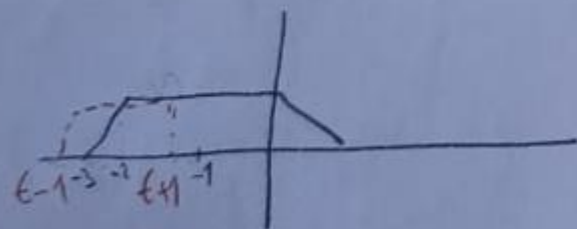


$$y(t) = \int_{-3}^{t+1} (\tau+3) \cdot 1 d\tau = \frac{\tau^2}{2} + 3\tau \Big|_{-3}^{t+1}$$

$$= \frac{(t+1)^2}{2} + 3(t+1) - \left(\frac{3^2}{2} + 9 \right)$$

$$= \frac{t^2}{2} + 4t + 8$$

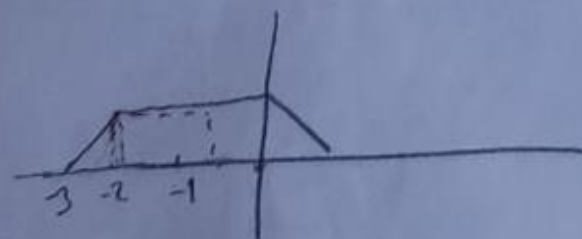
$$-3 \leq t < -2$$



$$y(t) = \int_{-3}^{-2} (\tau+3) \cdot 1 d\tau + \int_{-2}^{t+1} 1 \cdot 1 d\tau = \frac{\tau^2}{2} + 3\tau \Big|_{-3}^{-2} + \tau \Big|_{-2}^{t+1}$$

$$= t + \frac{7}{2}$$

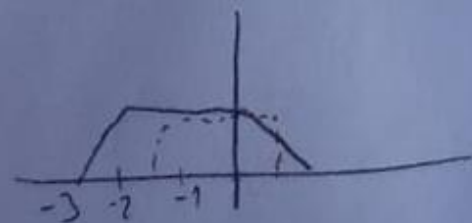
$$-2 \leq t < -1$$



$$y(t) = \int_{t-1}^{-2} (\tau+3) d\tau + \int_{-2}^{t+1} 1 \cdot 1 d\tau = \frac{\tau^2}{2} + 3\tau \Big|_{t-1}^{-2} + \tau \Big|_{-2}^{t+1}$$

$$= -\frac{t^2}{2} - t + \frac{3}{2}$$

$$-1 \leq t < 0$$



$$y(t) = \int_{t-1}^0 1 \cdot 1 d\tau + \int_0^{t+1} (1-\tau) d\tau = \tau \Big|_{t-1}^0 + \tau - \frac{\tau^2}{2} \Big|_0^{t+1}$$

$$= -\frac{t^2}{2} - t + \frac{3}{2}$$

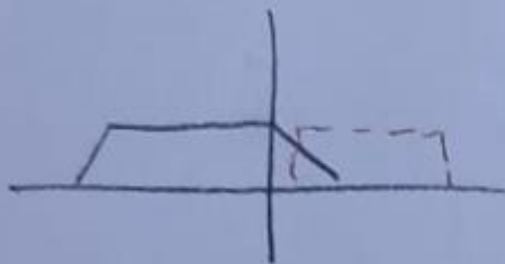
$$0 \leq t < 1$$



$$y(t) = \int_{t-1}^0 1 \cdot 1 \, d\tau + \int_0^1 (1-\tau) \, d\tau = \tau \Big|_{t-1}^0 + \tau - \frac{\tau^2}{2} \Big|_0^1$$

$$= -t + \frac{3}{2}$$

$$1 \leq t < 2$$



$$y(t) = \int_{t-1}^1 (1-\tau) \, d\tau = \tau - \frac{\tau^2}{2} \Big|_{t-1}^1$$

$$= \frac{t^2}{2} - 2t + 2$$

$$t \geq 2 \quad y(t) = 0$$

$$y(t) = \begin{cases} 0 & t \leq -4 \text{ or } t \geq 2 \\ \frac{t^2}{2} + 4t + 8 & -4 \leq t < -3 \\ t + \frac{7}{2} & -3 \leq t < -2 \\ -\frac{t^2}{2} - t + \frac{3}{2} & -2 \leq t < -1 \\ -\frac{t^2}{2} - t + \frac{3}{2} & -1 \leq t < 0 \\ -t + \frac{3}{2} & 0 \leq t < 1 \\ \frac{t^2}{2} - 2t + 2 & 1 \leq t < 2 \end{cases}$$